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1.0 INTRODUCTION

Diabetes is a chronic disease that impacts the way the body metabolizes blood sugar and it continues to grow in prevalence worldwide. Prevention and management of diabetes are most effective with a clear picture of the relationships between different factors. This report takes its foundation from a dataset comprising a series of health variables. From this data, we hope to obtain useful information on trends and relationships ruling diabetes and its linked health outcomes. Table 1.1 shows the column attributes and their descriptions that were involved in the dataset.

Table 1.1 Column attributes and its description

Attribute (Column Name)	Description
Age	Age of the individual (in years)
Sex	Biological sex (Male or Female)
Ethnicity	Ethnic background (e.g., White, Asian, etc.)
BMI	Body Mass Index (weight/height ²), indicates weight category
Waist_Circumference	Waist measurement in centimeters
Fasting_Blood_Glucose	Blood sugar level after fasting (mg/dL)
HbA1c	Long-term average blood sugar level (%)
Blood_Pressure_Systolic	Top number in blood sugar level (%)
Blood_Pressure_Diastolic	Bottom number in blood pressure reading (mmHg)
Cholesterol_Total	Total cholesterol level (mg/dL)
Cholesterol_HDL	“Good” cholesterol (High-Density Lipoprotein)
Cholesterol_LDL	“Bad” cholesterol (Low-Density Lipoprotein)
GGT	Liver enzyme Gamma-Glutamyl Transferase (U/L)
Serum_Uterate	Uric acid level in blood (mg/dL)
Physical_Activity_Level	Level of physical activity (Low, Moderate, High)
Dietary_Intake_Calories	Estimated daily caloric intake
Alcohol_Consumption	Alcohol consumption level (None, Moderate, Heavy)

Smoking_Status	Smoking habit (Never, Former, Current)
Family_History_of_Diabetes	Whether the person has family history of diabetes (1 = Yes, 0 = No)
Previous_Gestational_Diabetes	Whether a woman had diabetes during pregnancy (1 = Yes, 0 = No)

1.0 Business Insights

- 1.1. Moderate physical activity is linked to better blood sugar control
 - People with moderate activity have the lowest average HbA1c.
- 1.2. Males have a slightly higher BMI than females.
 - BMI : Males (29.54) vs. Females (29.30). Tailored health programs may be helpful.
- 1.3. Heavy drinkers show higher liver stress.
 - CGT enzyme is highest among heavy drinkers (56.01 U/L).
- 1.4. Family history has limited impact on fasting glucose in this dataset.
 - Average glucose : With history (134.75), without history (134.80).
- 1.5. Women with previous gestational diabetes average 44 years old.
 - Early monitoring post-pregnancy is key to preventing type 2 diabetes.
- 1.6. Smoking status has minimal effect on cholesterol level.
 - Cholesterol is nearly the same across current, former, and never smokers.

2.0 Characteristic on business insight and the collected dataset

2.1. Moderate physical activity is linked to better blood sugar control

Dataset Characteristic:

- The dataset includes a *Physical_Activity_Level* column with categories like *Low*, *Moderate* and *High*.
- It also tracks *HbA1c* (blood sugar level) levels for each person.
- By grouping *HbA1c* by activity level, we see moderate activity consistently shows the lowest average, suggesting physical activity is recorded well and affects blood sugar in this population.

2.2. Males have a slightly higher BMI than females.

Dataset Characteristic:

- The dataset includes *Sex* column and a *BMI* column.
- This allows easy comparison across gender.
- The dataset shows males average 29.54 BMI, females 29.30, confirming that gender-based health differences are captured and measurable.

2.3. Heavy drinkers show higher liver stress.

Dataset Characteristic:

- *Alcohol_consumption* is categorized (e.g., Non, Moderate, Heavy).
- *GGT* (Gamma-Glutamyl Transferase) is a liver enzyme included in the dataset.
- This allows liver health to be compared across drinking habits and shows the dataset can connect lifestyle to organ stress.

2.4. Family history has limited impact on fasting glucose

Dataset Characteristic:

- The dataset includes *Family_of_Diabetes* column and *Fasting_Blood_Glucose*
- This shows the dataset captures both hereditary and medical lab data
- This suggest lifestyle play a bigger role

2.5. Women with previous gestational diabetes average 44 years old

Dataset Characteristic:

- This includes a *Previous_Gestational_Diabetes* column and *Age*

- This helps identify and isolate this subgroup to study their health
- It reflects valuable info for early diabetes risk in women

2.6. Smoking status has minimal effect on cholesterol level

Dataset Characteristic:

- Smoking_status is a clear, multi class column
- Cholesterol_Total is a numeric lab result available for all patients
- This shows the dataset allows lifestyle-health comparisons but does not always find strong effects

CONCLUSION

The findings contain important diabetes health implications. Enhanced control of blood glucose relates to moderate physical activity and variations in BMI by gender point to opportunities for intervention targeting. High levels of alcohol drinking are linked to added liver stress and family history heredity shows little impact on fasting blood glucose levels, pointing towards a lifestyle determinant. Former gestational diabetic women mean age is 44 years which means normal follow-up needs to be regular. Smoking has minimal effect on cholesterol. Overall, the dataset supports targeted health strategies based on lifestyle and demographic factors.